SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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Abstract.

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1. Introduction

1.1. Background and motivation.

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$ and let G be the corresponding finite reflection group. Let $k : R \to \mathbb{R}_+$ be G-invariant multiplicity function. Let $R_{\mathrm{Dunkl},j}$ be the j-th Dunkl-Riesz transform corresponding to R and k. Let w be the Dunkl weight corresponding to R and k. Let $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N)$.*

(i) *If $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p > N$, then $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))} \leq c_{p,R,k}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

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(ii) *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, $p > N$, *then* $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))} \leq c_{p,R,k}^{-1}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

(iii) *If* $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, *then* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \leq c_{R,k}\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

(iv) *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, *then* $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \geq c_{R,k}^{-1}\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

1.2. **Strategy of the proof.** In Section 3, we introduce our *first ad hoc* Sobolev space and demonstrate that its semi-norm is controlled by the usual Sobolev semi-norm. In Section 4, we use Janson-Wolff method to control the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$ by the *first ad hoc* Sobolev semi-norm of f. Thus, we control the $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$ by the usual Sobolev semi-norm (i.e., we prove Theorem 1.1 (i)).

In Section 5, we control, in every chamber Γ, the Sobolev semi-norm of f via $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$. This is done by verifying the assumptions in Theorem 1.4 in [6] and applying the latter theorem.

In Section 6, we introduce our *second ad hoc* Sobolev space and use the Rochberg-Semmes method to demonstrate that the semi-norm of the latter space is controlled by $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$.

In Section 7, for every face of each chamber, we compare the boundary traces of f taken from the adjacent regions: the *interior trace* is computed by approaching the face from within the given chamber, while the *exterior trace* is obtained by approaching the same face from the neighboring chamber that shares the same face. Those boundary values are demonstrated to be the same. This allows us to show that the function f is globally Sobolev and that its Sobolev semi-norm is controlled by $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))}$-norm of the commutator $[R_{\mathrm{Dunkl},j}, M_f]$ (i.e., we prove Theorem 1.1 (ii)).

1.3. **Notation.** Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N, p, and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda\,\ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p, which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ-measurable set E by

$$(f)_E := \fint_E f(x)d\mu(x) := \frac{1}{\mu(E)}\int_E f(x)d\mu(x). \tag{1.1}$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. Preliminaries

2.1. **Harmonic analysis in the Dunkl setting.** Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\| \cdot \|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v\rangle v.$$

A finite set $R \subset \mathbb{R}^N \backslash \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $|v| = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v\in R}$ is called the Weyl group (reflection group) of the root system R. Let $k : R \to [0, \infty)$ be a multiplicity function, which is a G-invariant function fixed throughout the paper.

Define the Dunkl measure $w(x)dx$ on $\mathbb{R}^N$ by

$$w(x) = \prod_{v\in R} |\langle x, v\rangle|^{k(v)}.$$

Let $B(x, r) := \{y \in \mathbb{R}^N : \ |x - y| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Furthermore,

$$w(B(x, r)) \sim r^N \prod_{v\in R} (|\langle x, v\rangle| + r)^{k(v)}, \tag{2.1}$$

where the implicit constant only depends on N, R and k. It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$C^{-1} \left(\frac{r_2}{r_1}\right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1}\right)^{N'}, \text{ for } 0 < r_1 < r_2, \tag{2.2}$$

so the numbers N' and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v\in R} \frac{k(v)}{2} \langle v, \xi\rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x\rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ. Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v\in R} k(v)\delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x\rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x\rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \ldots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$\left\{x \in \mathbb{R}^N : \langle x, v\rangle \neq 0 \ \ \textit{for all } v \in R\right\}. \tag{2.3}$$

Define

$$d(x,y)=\min_{\sigma\in G}|x-\sigma(y)|,\quad x,y\in\mathbb{R}^N.$$

Recall from [5, Chapter VII, proof of Theorem 2.12] that

$$d(x,y)=|x-y| \text{ if and only if } x,y\in\Gamma \text{ for some chamber } \Gamma. \tag{2.4}$$

Denote by $K_{\mathrm{Dunkl},j}(x,y)$ the integral kernel of $R_{\mathrm{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [4]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [4, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j\in\{1,2,\cdots,N\}$ and every x,y with $d(x,y)\neq 0$,*

$$|K_{\mathrm{Dunkl},j}(x,y)|\le\frac{d(x,y)}{|x-y|}\cdot\frac{C}{w(B(x,d(x,y)))}.$$

Proof. The proof is given in [4, Theorem 1.1]. □

Lemma 2.3. *We have*

$$d(x,y)^2\cdot w(x)w(y)\le Cw^2(B(x,d(x,y))),\quad x,y\in\mathbb{R}^N.$$

Proof. Define equivalence relation $\sim$ on R by saying $v_1\sim v_2$ if there exists $g\in G$ with $v_1=gv_2$. Let us split R into equivalence classes with respect to $\sim$. Since k is G-invariant, it follows that k is constant on each equivalence class. Note that G leaves each equivalence class invariant.

We, therefore, write

$$w(x)=\prod_{A\in R/\sim}\Big(\prod_{v\in A}|\langle x,v\rangle|\Big)^{k|_A}.$$

On the other hand,

$$w(B(x,r))\ge c^{(1)}_{R,k}r^N\prod_{A\in R/\sim}\Big(\prod_{v\in A}(r+|\langle x,v\rangle|)\Big)^{k|_A}.$$

Let $x\in\Gamma$ and $y\in g\Gamma$. Set $z=g^{-1}y\in\Gamma$. Note that $d(x,y)=|x-z|$ and, therefore,

$$|\langle x,v\rangle|+d(x,y)=|\langle x,v\rangle|+|x-z|\ge 2^{-\frac12}|\langle z,v\rangle|.$$

Thus,

$$\prod_{v\in A}(|\langle x,v\rangle|+d(x,y))\ge\prod_{v\in A}2^{-\frac12}|\langle z,v\rangle|=2^{-\frac12|A|}\prod_{v\in A}|\langle g^{-1}y,v\rangle|=$$
$$=2^{-\frac12|A|}\prod_{v\in A}|\langle y,gv\rangle|=2^{-\frac12|A|}\prod_{v\in gA}|\langle y,v\rangle|=2^{-\frac12|A|}\prod_{v\in A}|\langle y,v\rangle|.$$

Thus,

$$w(B(x,d(x,y)))\ge c^{(1)}_{R,k}d^N(x,y)\prod_{A\in R/\sim}\Big(\prod_{v\in A}(d(x,y)+|\langle x,v\rangle|)\Big)^{k|_A}\ge$$
$$\ge c^{(1)}_{R,k}d^N(x,y)\prod_{A\in R/\sim}\Big(2^{-\frac12|A|}\prod_{v\in A}|\langle y,v\rangle|\Big)^{k|_A}=c^{(2)}_{R,k}d^N(x,y)w(y).$$

On the other hand,

$$w(B(x,d(x,y)))\ge c^{(1)}_{R,k}d^N(x,y)w(x).$$

Multiplying the last two equations, we complete the proof. □

Lemma 2.4. *Let $1 \le j \le N$. If $x_j \ge y_j$, then*

$$K_{\mathrm{Dunkl},j}(x,y) \ge c_{R,k} \frac{x_j - y_j}{|x-y|} \cdot \frac{1}{w(B(x,|x-y|))}.$$

If $x_j \le y_j$, then

$$K_{\mathrm{Dunkl},j}(x,y) \le c_{R,k} \frac{x_j - y_j}{|x-y|} \cdot \frac{1}{w(B(x,|x-y|))}.$$

Proof. The lower bound is identical to that in [4, Theorem 1.2]. □

2.2. **Schatten class.** [Zhijie will add this later]

2.3. **Nearly weakly orthonormal sequences.** Rochberg and Semmes [11] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [3, 7, 10, 1, 2, 9, 11, 13, 12]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [7, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Definition 2.5. *Let $\mathcal{D}$ be the dyadic lattice in $\mathbb{R}^N$. A sequence of functions $\{e_Q\}_{Q\in\mathcal{D}}$ is called nearly weakly orthonormal (NWO, in short) sequence if*

(i) $\operatorname{supp}(e_Q) \subset cQ$ *for some positive constant c independent of Q.*
(ii) $\|e_Q\|_{L_\infty(\mathbb{R}^N, w(x)dx)} \le w(Q)^{-\frac{1}{2}}$.

Lemma 2.6. *Let $1 < p < \infty$ and let $\mathcal{D}$ be the dyadic lattice in $\mathbb{R}^N$. If $T \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, then*

$$\left\| \left\{ \langle Te_Q, f_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q\in\mathcal{D}} \right\|_{l_p(\mathcal{D})} \le c_{N,p,w,c_1,c_2} \|T\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$$

for every NWO systems $\{e_Q\}_{Q\in\mathcal{D}}$ and $\{f_Q\}_{Q\in\mathcal{D}}$ (with respective constants c_1 and c_2).

Proof. reference to [11] is inappropriate. Rochberg and Semmes do not consider weighted setting. we absolutely must provide correct reference. This property can be found in [11, (1.10)] (see also [7, Section 7]). □

3. Comparison of Sobolev norms

Theorem 3.1. *For every $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p > N$, we have*

$$\left(\int_{\mathbb{R}^N\times\mathbb{R}^N} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x)-f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}} \le c_{p,R} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Lemma 3.2. *For every convex polyhedral cone Γ, we have*

$$\int_\Gamma \frac{dx}{|x-y|^{2N}} \ge c_\Gamma \operatorname{dist}(y,\Gamma)^{-N}, \quad y \notin \Gamma.$$

Proof. Find a point $z \in \Gamma$ such that

$$|y-z| = \mathrm{dist}(y, \Gamma).$$

We have $z + \Gamma \subset \Gamma$. Thus,

$$\int_\Gamma \frac{dx}{|x-y|^{2N}} \geq \int_{z+\Gamma} \frac{dx}{|x-y|^{2N}} = \int_\Gamma \frac{dx}{|x-(y-z)|^{2N}}.$$

Clearly,

$$|x-(y-z)| \leq |x| + |y-z|.$$

Thus,

$$\int_\Gamma \frac{dx}{|x-y|^{2N}} \geq \int_\Gamma \frac{dx}{(|x|+|y-z|)^{2N}} = |y-z|^{-N} \int_\Gamma \frac{dx}{(|x|+1)^{2N}}.$$

□

Lemma 3.3. *If $p \in (N, 2N)$, then*

$$\sup_{a>0} \int_{\mathbb{R}^N} \frac{|x-ae_N|^{p-2N} dx}{(1+|x|)^p} < \infty.$$

Proof. **Step 1:** we claim that, for every $a > 0$,

$$\int_{\mathbb{R}^N} \frac{|x-ae_N|^{p-2N} dx}{(1+|x|)^p} \leq$$

$$\leq 4^p \int_{\mathbb{R}^{N-1}\times(-a,a)} \frac{|x|^{p-2N} dx}{(1+|x'|+a)^p} + 3^{2N-p} \int_{\mathbb{R}^N} \frac{|x|^{p-2N} dx}{(1+|x|)^p}.$$

We clearly have

$$\int_{\mathbb{R}^N} \frac{|x-ae_N|^{p-2N} dx}{(1+|x|)^p} =$$

$$= \int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N} dx}{(1+|x|)^p} + \int_{\mathbb{R}^{N-1}\times\mathbb{R}\setminus(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N} dx}{(1+|x|)^p}.$$

If $x_N \notin (\frac{a}{2}, \frac{3a}{2})$, then $|x_N - a| \geq \frac{1}{3}|x_N|$. Hence, $|x - ae_N| \geq \frac{1}{3}|x|$. Thus,

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}\setminus(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N} dx}{(1+|x|)^p} \leq$$

$$\leq \int_{\mathbb{R}^{N-1}\times\mathbb{R}\setminus(\frac{a}{2},\frac{3a}{2})} \frac{3^{2N-p}|x|^{p-2N} dx}{(1+|x|)^p} \leq \int_{\mathbb{R}^N} \frac{3^{2N-p}|x|^{p-2N} dx}{(1+|x|)^p}.$$

If $x_N \in (\frac{a}{2}, \frac{3a}{2})$, then (here, we using the notation $x' = (x_1, \cdots, x_{N-1})$)

$$1 + |x| \geq 1 + \max\{|x'|, \frac{1}{2}a\} \geq \frac{1}{4}(1 + |x'| + a).$$

Thus,

$$\int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N} dx}{(1+|x|)^p} \leq 4^p \int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N} dx}{(1+|x'|+a)^p} =$$

$$= 4^p \int_{\mathbb{R}^{N-1}\times(-\frac{a}{2},\frac{a}{2})} \frac{|x|^{p-2N} dx}{(1+|x'|+a)^p} \leq 4^p \int_{\mathbb{R}^{N-1}\times(-a,a)} \frac{|x|^{p-2N} dx}{(1+|x'|+a)^p}.$$

Combining all preceding estimates, we infer the claim in Step 1.

Step 2: we claim that

$$\sup_{a>0}\int_{\mathbb{R}^{N-1}\times(-a,a)}\frac{|x|^{p-2N}dx}{(1+|x'|+a)^p}<\infty.$$

Passing to polar coordinates in $\mathbb{R}^{N-1}$, we equivalently rewrite the claim as

$$\sup_{a>0}\int_{\mathbb{R}_+\times(0,a)}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(1+r+a)^p}<\infty.$$

If $0<a<1$, then

$$\int_{\mathbb{R}_+\times(0,a)}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(1+r+a)^p}\leq\int_{\mathbb{R}_+\times(0,1)}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(1+r)^p}.$$

If $a\geq 1$, then

$$\int_{\mathbb{R}_+\times(0,a)}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(1+r+a)^p}\leq\int_{\mathbb{R}_+\times(0,a)}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(r+a)^p}=$$

$$=\int_{(a,\infty)\times(0,a)}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(r+a)^p}+$$

$$+\int_0^a\int_0^r\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}dx_Ndr}{(r+a)^p}+\int_0^a\int_0^{x_N}\frac{(r^2+x_N^2)^{\frac{p}{2}-N}r^{N-2}drdx_N}{(r+a)^p}\leq$$

$$\leq{}^{1}\int_{(a,\infty)\times(0,a)}\frac{r^{p-2N}r^{N-2}drdx_N}{r^p}+$$

$$+\int_0^a\int_0^r\frac{r^{p-2N}r^{N-2}dx_Ndr}{a^p}+\int_0^a\int_0^{x_N}\frac{x_N^{p-2N}r^{N-2}drdx_N}{a^p}=$$

$$=\int_a^\infty r^{-N-1}dr+a^{-p}\int_0^a r^{p-N-1}dr+\frac{1}{N-1}a^{-p}\int_0^a x_N^{p-N-1}dx_N=$$

$$=\frac{1}{N}a^{-N}+\frac{1}{p-N}a^{-N}+\frac{1}{(N-1)(p-N)}a^{-N}\leq\frac{1}{N}+\frac{1}{p-N}+\frac{1}{(N-1)(p-N)}.$$

This yields the claim in Step 2.

Combining Steps 1 and 2, we complete the proof of the lemma. □

Lemma 3.4. *If $p\in(N,2N)$, then*

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}_+}\frac{|x-y|^{p-2N}dx}{|x+z|^p}\leq c_{p,N}z_N^{-N},\quad y,z\in\mathbb{R}^{N-1}\times\mathbb{R}_+.$$

Proof. By dilation invariance, we may assume without loss of generality that $z_N=1$. By translation invariance in $\mathbb{R}^{N-1}$, we may assume without loss of generality that $z_1=\cdots=z_{N-1}=0$. We, therefore, have

$$|x+z|=|x+e_N|\geq\max\{1,|x|\}\geq\frac{1}{2}(1+|x|),\quad x\in\mathbb{R}^{N-1}\times\mathbb{R}_+.$$

Thus,

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}_+}\frac{|x-y|^{p-2N}dx}{|x+z|^p}\leq 2^p\int_{\mathbb{R}^{N-1}\times\mathbb{R}_+}\frac{|x-y|^{p-2N}dx}{(1+|x|)^p}\leq$$

$$=2^p\int_{\mathbb{R}^N}\frac{|x-y|^{p-2N}dx}{(1+|x|)^p}=2^p\int_{\mathbb{R}^N}\frac{\big|x-|y|e_N\big|^{p-2N}dx}{(1+|x|)^p}.$$

[1] It is exactly at this point we are using the assumption $p<2N$.

The assertion follows from Lemma 3.3. □

We need the following elementary version of the Hahn-Banach theorem.

Lemma 3.5. *Let $K \subset \mathbb{R}^N$ be a convex set and let $y \notin K$. Let $z \in K$ be the point closest to y. Affine hyperplane P passing through z and orthogonal to $y - z$ splits the space into two half-spaces, one of them containing y, the other one containing K.*

Proof. By translation invariance, we may assume without loss of generality that $z = 0$. By rotation and dilation invariance, we may assume without loss of generality that $y = e_N$. By assumption, $\mathrm{dist}(e_N, K) = 1$. We need to show that K is contained in the (closed) lower half-plane.

Assume the contrary: let $x \in K$ be such that $x_N > 0$. Since $0 \in K$ and since K is convex, it follows that $\epsilon x \in K$ for every $\epsilon \in (0, 1)$. Thus,

$$\mathrm{dist}(e_N, K) \leq |e_N - \epsilon x|, \quad \epsilon \in (0, 1).$$

Since $\epsilon \in (0, 1)$ is arbitrary, it follows that

$$\mathrm{dist}(e_N, K) \leq \inf_{\epsilon \in (0,1)} |e_N - \epsilon x| = \inf \big(\epsilon^2 \sum_{k=1}^{N-1} x_k^2 + (1 - \epsilon x_N)^2\big)^{\frac{1}{2}}.$$

Since $x_N > 0$, it follows that infimum is strictly less than 1. Thus, $\mathrm{dist}(e_N, K) < 1$. However, it is granted in the lemma that $\mathrm{dist}(e_N, K) = 1$. This contradiction shows that such an x cannot exist. This completes the proof. □

Lemma 3.6. *If $p \in (N, 2N)$, then*

$$\int_\Gamma \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq c_p \mathrm{dist}(gy, \Gamma)^{-N}, \quad y \in \Gamma.$$

Proof. Find a point $z \in \Gamma$ such that

$$|gy - z| = \mathrm{dist}(gy, \Gamma).$$

By Lemma 3.5, affine hyperplane passing through z and orthogonal to $gy - z$ splits the space into two half-spaces, one of them containing gy, the other one containing Γ (denote this one by H).

We have $y \in H$, $\Gamma \subset H$ and $gy \notin H$. So,

$$\int_\Gamma \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq \int_H \frac{|x - y|^{p-2N} dx}{|x - gy|^p}.$$

Using Lemma 3.4, we write

$$\int_H \frac{|x - y|^{p-2N} dx}{|x - gy|^p} \leq c_{p,N} \mathrm{dist}(gy, H)^{-N}.$$

Since $z \in \partial H$ and since $gy - z$ is orthogonal to ∂H, it follows that $\mathrm{dist}(gy, H) = |gy - z| = \mathrm{dist}(gy, \Gamma)$. □

Proof of Theorem 3.1. **Case 1:** Let $p \in (N, 2N)$.

We have

$$\Big(\int_{\Gamma \times g\Gamma} \Big(\frac{d(x, y)}{|x - y|}\Big)^p |f(x) - f(y)|^p \frac{dxdy}{d(x, y)^{2N}}\Big)^{\frac{1}{p}} =$$

$$= \Big(\int_{\Gamma \times \Gamma} \Big(\frac{|x - y|}{|x - gy|}\Big)^p |f(x) - f(gy)|^p \frac{dxdy}{|x - y|^{2N}}\Big)^{\frac{1}{p}} \leq$$

$$\leq \Big(\int_{\Gamma\times\Gamma}\Big(\frac{|x-y|}{|x-gy|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}+$$

$$+\Big(\int_{\Gamma\times\Gamma}\Big(\frac{|x-y|}{|x-gy|}\Big)^p|f(y)-f(gy)|^p\frac{dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}\leq$$

$$\leq \Big(\int_{\Gamma\times\Gamma}|f(x)-f(y)|^p\frac{dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}+$$

$$+\Big(\int_{\Gamma}|f(y)-f(gy)|^p\Big(\int_{\Gamma}\frac{|x-y|^{p-2N}}{|x-gy|^p}dx\Big)dy\Big)^{\frac{1}{p}}.$$

Obviously, the first summand on the right hand side is less than $\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$. By Lemmas 3.6 and 3.2,

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+$$

$$+c^{(1)}_{p,R}\Big(\int_{\Gamma}|f(y)-f(gy)|^p\Big(\int_{\Gamma}\frac{dx}{|x-gy|^{2N}}\Big)dy\Big)^{\frac{1}{p}}=$$

$$=\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+c^{\frac{1}{p}}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(y)-f(gy)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}\leq$$

$$\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+c^{(1)}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(gy)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}+$$

$$+c^{(1)}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}.$$

Now,

$$\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(gy)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}=\Big(\int_{\Gamma\times g\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$$

and

$$\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}\leq\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^pdxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}\leq\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Thus,

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq(1+2c^{(1)}_{p,R})\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Summing over Γ and $1\neq g\in G$, we obtain

$$\Big(\int_{\mathbb{R}^N\times\mathbb{R}^N}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq|G|^2(1+2c^{(1)}_{p,R})\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Case 2: Let $p\geq 2N$. We have

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq$$

$$\leq\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^{2N}|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}=\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

□

4. Proof of Theorem 1.1 (i)

Let (X,μ) be a measure space and let $L_p(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)$ denote the mixed-norm space consisting of all $\mu\times\mu$-measurable functions $f:X\times X\to\mathbb{C}$ such that

$$\int_X \|f(x,\cdot)\|^p_{L_{q,\infty}(X,\mu)}d\mu(x),\ \int_X \|f(\cdot,y)\|^p_{L_{q,\infty}(X,\mu)}d\mu(y)<\infty.$$

We equip $L_p(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)$ with its natural norm.

Lemma 4.1. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $K\in L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)$. The integral operator*

$$T_Kf(x):=\int_X K(x,y)f(y)d\mu(y)$$

belongs to $S_{p,\infty}(L_2(X,d\mu))$. Moreover, we have

$$\|T_K\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|K\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Proof. The proof is given in [8, Lemma 2] (see also [7, Corollary 11.2]). □

Lemma 4.2. *Let $p>2$ and let $\frac{1}{q}=1-\frac{2}{p}$. We have*

$$\|F_1F_2\|_{L_p(L_{p',\infty})(X\times X,\mu\times\mu)}\le 2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$

Proof. We clearly have

$$LHS=\Big(\int_X \|F_1(x,\cdot)F_2(x,\cdot)\|^p_{L_{p',\infty}(X,\mu)}d\mu(x)\Big)^{\frac{1}{p}}.$$

Clearly, $\frac{1}{p'}=\frac{1}{p}+\frac{1}{q}$. Thus,

$$\|F_1(x,\cdot)F_2(x,\cdot)\|_{L_{p',\infty}(X,\mu)}\le 2^{\frac{1}{p'}}\|F_1(x,\cdot)\|_{L_{p,\infty}(X,\mu)}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}\le$$
$$\le 2^{\frac{1}{p'}}\|F_1(x,\cdot)\|_{L_p(X,\mu)}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}.$$

Hence,

$$LHS\le 2^{\frac{1}{pp'}}\Big(\int_X \|F_1(x,\cdot)\|^p_{L_p(X,\mu)}\|F_2(x,\cdot)\|^p_{L_{q,\infty}(X,\mu)}d\mu(x)\Big)^{\frac{1}{p}}\le$$
$$\le 2^{\frac{1}{pp'}}\Big(\int_X \|F_1(x,\cdot)\|^p_{L_p(X,\mu)}d\mu(x)\Big)^{\frac{1}{p}}\times\sup_{x\in X}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}=$$
$$=2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$

□

Lemma 4.3. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $F\in L_p(X\times X,H^2(\mu\times\mu))$ and $H\in L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)$. We have*

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{1-\frac{2}{p}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Proof. By Lemma 4.1,

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|FH\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Let $\frac{1}{q}=1-\frac{2}{p}$. It follows from Lemma 4.2 that

$$\|F_1F_2\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)}\le$$

$$\leq 2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Setting $F_1 = FH^{\frac{2}{p}}$ and $F_2 = H^{\frac{1}{q}}$, we obtain

$$\|FH\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)} \leq$$

$$\leq 2^{\frac{1}{pp'}}\|FH^{\frac{2}{p}}\|_{L_p(X\times X,\mu\times\mu)}\|H^{\frac{1}{q}}\|_{L_\infty(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)} =$$

$$= 2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{\frac{1}{q}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)} =$$

$$= 2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{1-\frac{2}{p}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

□

Lemma 4.4. *Let (X,μ) be a measure space and let $2 < p < \infty$. Suppose that $F \in L_p(X\times X, GH^2(\mu\times\mu))$ and $H \in L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)$. We have*

$$\|T_{FH}\|_{S_p(L_2(X,d\mu))} \leq c_{p,H}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}.$$

Proof. The assertion follows from Lemma 4.3 by real interpolation. □

The following proposition, strictly speaking, is not part of the proof of Theorem 1.1 (i). Instead, it features in the proof of Theorem 1.1 (ii) in the next section. However, in terms of the employed technique, it certainly belongs to this section.

Proposition 4.5. *Let $p > 2$, let Γ be a chamber and let $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$. Let V be an integral operator whose integral kernel K satisfies the condition*

$$|K(x,y)| \leq \frac{1}{w(B(x,|x-y|))}.$$

For every $1 \leq j \leq N$, we have

$$\|[V,M_f]\|_{\mathcal{L}_p(L_2(\Gamma,w(x)dx))} \leq c_{R,k}\|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)}.$$

Proof. In what follows, let $X = \Gamma$ and $d\mu(x) = w(x)dx$. Set

$$F(x,y) = w(B(x,|x-y|))K(x,y)(f(x)-f(y)), \quad x \neq y,$$

$$H(x,y) = \frac{1}{w(B(x,|x-y|))}, \quad x \neq y.$$

By [7, Lemma 5.7], $H \in L_\infty(L_{1,\infty})^{\rm symm}(\Gamma\times\Gamma,\mu\times\mu)$. By assumption on the kernel K,

$$\|F\|_{L_p(\Gamma\times\Gamma,G^2(\mu\times\mu))} \leq \Big(\int_{\Gamma\times\Gamma}|f(x)-f(y)|^p\frac{w(x)w(y)dxdy}{w^2(B(x,|x-y|))}\Big)^{\frac{1}{p}}.$$

By (2.1),

$$w(B(x,|x-y|)) \geq c'_{R,k}|x-y|^N w^{\frac{1}{2}}(x)w^{\frac{1}{2}}(y), \quad x,y \in \mathbb{R}^N.$$

Thus,

$$\|F\|_{L_p(\Gamma\times\Gamma,H^2(\mu\times\mu))} \leq$$

$$\leq (c'_{R,k})^{-\frac{2}{p}}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^p}{|x-y|^{2N}}dxdy\Big)^{\frac{1}{p}} = (c'_{R,k})^{-\frac{2}{p}}\|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)}.$$

Obviously, the integral kernel of the operator $[V,M_f]$ is FH. The assertion follows now from Lemma 4.4. □

Proof of Theorem 1.1 (i). In what follows, let $X = \mathbb{R}^N$ and $d\mu(x) = w(x)dx$. Set

$$F(x,y) = w(B(x,d(x,y)))K_{\mathrm{Dunkl},j}(x,y)(f(x)-f(y)), \quad x \neq y,$$

$$H(x,y) = \frac{1}{w(B(x,d(x,y)))}, \quad x \neq y.$$

By [7, Lemma 11.7], $H \in L_\infty(L_{1,\infty})^{\mathrm{symm}}(X \times X, \mu \times \mu)$. By Lemma 2.2,

$$\|F\|_{L_p(X\times X, H^2(\mu\times\mu))} \leq$$

$$\leq c^{(1)}_{R,k}\left(\int_{\mathbb{R}^N\times\mathbb{R}^N}\left(\frac{d(x,y)}{|x-y|}\right)^p |f(x)-f(y)|^p \frac{w(x)w(y)dxdy}{w(B(x,d(x,y)))^2}\right)^{\frac{1}{p}}.$$

Using Lemma 2.3, we conclude that

$$\|F\|_{L_p(X\times X, H^2(\mu\times\mu))} \leq c^{(2)}_{R,k}\left(\int_{\mathbb{R}^N\times\mathbb{R}^N}\left(\frac{d(x,y)}{|x-y|}\right)^p |f(x)-f(y)|^p \frac{dxdy}{d(x,y)^{2N}}\right)^{\frac{1}{p}}.$$

Using Theorem 3.1, we conclude that

$$\|F\|_{L_p(X\times X, H^2(\mu\times\mu))} \leq c_{R,k}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Obviously, the integral kernel of the operator $[R_{\mathrm{Dunkl},j}, M_f]$ is FH. The assertion follows now from Lemma 4.4. □

5. Proof of restricted version of Theorem 1.1 (ii)

Theorem 5.1. *If $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$ for every chamber Γ. Furthermore,*

$$\|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)} \leq c_{R,k}\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}.$$

Lemma 5.2. *Let Γ be a chamber and let $x_0 \in \Gamma$. We have*

$$\sup_{x\in B(x_0,r)\cap\Gamma} w(x) \leq \frac{c_{k,\Gamma}}{m(B(x_0,r)\cap\Gamma)}\int_{B(x_0,r)\cap\Gamma} w(x)dx.$$

Proof. If $x = x_0 + y$, $y \in B(0,r)$, then

$$|\langle x, v\rangle| \leq |\langle x_0, v\rangle| + r|\langle y, v\rangle| \leq |\langle x_0, v\rangle| + r\sqrt{2}.$$

Thus,

$$w(x) = \prod_{v\in R}(|\langle x_0, v\rangle| + |\langle y, v\rangle|)^{\epsilon\cdot k(v)} \leq \prod_{v\in R}(|\langle x_0, v\rangle| + r\sqrt{2})^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v\in R}(|\langle x_0, v\rangle| + r\sqrt{2})^{k(v)}.$$

By convexity, $x_0 + \Gamma \subset \Gamma$. Hence,

$$x_0 + (B(0,r)\cap\Gamma) \subset B(x_0,r)\cap\Gamma.$$

So,

$$\int_{B(x_0,r)\cap\Gamma} w(x)dx \geq \int_{x_0+(B(0,r)\cap\Gamma)} w(x)dx = \int_{B(0,r)\cap\Gamma} w(x_0+y)dy.$$

If $y \in \Gamma$, then

$$|\langle x_0 + y, v\rangle| = |\langle x_0, v\rangle| + |\langle y, v\rangle|, \quad v \in R.$$

Thus,

$$w(x_0+y)=\prod_{v\in R}(|\langle x_0,v\rangle|+|\langle y,v\rangle|)^{k(v)}\geq$$
$$\geq\max\{\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)},\prod_{v\in R}|\langle y,v\rangle|^{k(v)}\}.$$

Hence,

$$\int_{B(x_0,r)\cap\Gamma}w(x)dx\geq\int_{B(0,r)\cap\Gamma}w(x_0+y)dy\geq$$
$$\geq\max\Big\{\int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)}dy,\int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle y,v\rangle|^{k(v)}dy\Big\}.$$

Obviously,

$$\int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)}dy=r^N\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)}\int_{B(0,1)\cap\Gamma}dy,$$
$$\int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle y,v\rangle|^{k(v)}dy=r^{N+\sum_{v\in R}k(v)}\int_{B(0,1)\cap\Gamma}\prod_{v\in R}|\langle y,v\rangle|^{k(v)}dy.$$

Thus,

$$\int_{B(x_0,r)\cap\Gamma}w(x)dx\geq c^{(1)}_{k,\Gamma}\max\{r^N\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)},r^{N+\sum_{v\in R}k(v)}\}.$$

Combining this with the preceding paragraph, we complete the proof. □

Lemma 5.3. *Let Γ be a chamber and let $x_0\in\Gamma$. Suppose $1<t<1+\frac{1}{|R|\cdot\max_{v\in R}k(v)}$. For every $r>0$, we have*

$$\Big(\frac{\int_{B(x_0,r)\cap\Gamma}w^{1-t}(x)dx}{\int_{B(x_0,r)\cap\Gamma}w(x)dx}\Big)^{\frac{1}{t}}\leq c_{t,k,\Gamma}\frac{m(B(x_0,r)\cap\Gamma)}{\int_{B(x_0,r)\cap\Gamma}w(x)dx}.$$

Proof. If $v\in R$ is such that

$$2\sqrt{2}r\leq|\langle x_0,v\rangle|,$$

then

$$|\langle x,v\rangle|\geq|\langle x_0,v\rangle|-|\langle x-x_0,v\rangle|\geq|\langle x_0,v\rangle|-r\sqrt{2}\geq\frac{1}{2}|\langle x_0,v\rangle|,\quad x\in B(x_0,r).$$

In this case,

$$\int_{B(x_0,r)}|\langle x,v\rangle|^{(1-t)|R|k(v)}dx\leq 2^{(t-1)|R|k(v)}|\langle x_0,v\rangle|^{(1-t)|R|k(v)}\cdot\int_{B(x_0,r)}dx=$$
$$=2^{(t-1)|R|k(v)}|\langle x_0,v\rangle|^{(1-t)|R|k(v)}\cdot\mathrm{Vol}(\mathbb{B}^N)r^N.$$

If $v\in R$ is such that

$$|\langle x_0,v\rangle|\leq 2\sqrt{2}r,$$

then let $y_0=\langle x_0,v\rangle$ and $z_0=x_0-\frac{1}{2}y_0v_0$. Note that $|y_0|\leq 2\sqrt{2}r$. Using new variables $y=\langle x,v\rangle$ and $z=x-\frac{1}{2}yv$, we write

$$\int_{B(x_0,r)}|\langle x,v\rangle|^{(1-t)|R|k(v)}dx=\int_{|y-y_0|^2+|z-z_0|^2\leq r^2}|y|^{(1-t)|R|k(v)}dydz\leq$$
$$\leq\int_{|y-y_0|^2,|z-z_0|^2\leq r^2}|y|^{(1-t)|R|k(v)}dydz=$$

$$= r^{N-1}\mathrm{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y-y_0|\le r} |y|^{(1-t)|R|k(v)} dy \le$$
$$= r^{N-1}\mathrm{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y|\le (1+2\sqrt{2})r} |y|^{(1-t)|R|k(v)} dy =$$
$$= r^{N+(1-t)|R|k(v)}\mathrm{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y|\le 1+2\sqrt{2}} |y|^{(1-t)|R|k(v)} dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_{B(x_0,r)} |\langle x,v\rangle|^{(1-t)|R|k(v)} dx \le c_1 R^n \big(\max\{|\langle x_0,v\rangle|, r\}\big)^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\int_{B(x_0,r)} w^{1-t}(x)dx \le \prod_{v\in R} \Big(\int_{B(x_0,r)} |\langle x,v\rangle|^{(1-t)|R|k(v)} dx\Big)^{\frac{1}{|R|}} \le$$
$$\le c_2 R^N \prod_{v\in R} \big(\max\{|\langle x_0,v\rangle|, r\}\big)^{(1-t)k(v)}.$$

The assertion follows easily. □

Lemma 5.4. *Let Γ be a chamber equipped with Euclidean distance. Measures dx and $w(x)dx$ and operator $T_{\Gamma,j} = M_{\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_\Gamma}$ satisfy on Γ the conditions in Theorem 1.4 in [6].*

Proof. **Step 1:** We claim that measures dx and $w(x)dx$ are A_∞-equivalent (in the sense of Proposition 1.2 in [6]). This verifies Assumption (i) in Theorem 1.4 in [6].

Let $x_0 \in \Gamma$. By Lemmas 5.2 and 5.3,

$$\Big(\frac{1}{m(B\cap\Gamma)}\int_{B\cap\Gamma} w^t(x)dx\Big)^{\frac{1}{t}} \le \frac{c_{k,\Gamma}}{m(B\cap\Gamma)}\int_{B\cap\Gamma} w(x)dx, \quad t>0.$$
$$\Big(\frac{\int_{B\cap\Gamma} w^{1-t}(x)dx}{\int_{B\cap\Gamma} w(x)dx}\Big)^{\frac{1}{t}} \le c_G \frac{m(B\cap\Gamma)}{\int_{B\cap\Gamma} w(x)dx}, \quad 1<t<1+\frac{1}{|R|\cdot\max_{v\in R}k(v)}.$$

The assertion follows now from Proposition 3.6 in [6].

Step 2: Assumption (ii) in Theorem 1.4 in [6] is obvious.

Step 3: Integral kernel K_j of the operator $T_{\Gamma,j}$ satisfies the estimates

$$|K_j(x,y)| \le \frac{c_G}{w(B(x,|x-y|)\cap\Gamma)}, \quad x,y\in\Gamma,$$
$$|K_j(x,y)-K_j(x',y)| \le \frac{|x-x'|}{|x-y|}\cdot\frac{c_G}{w(B(x,|x-y|))}, \quad |x-x'|\le\frac{1}{2}|x-y|, \quad x,x',y\in\Gamma,$$
$$|K_j(x,y)-K_j(x,y')| \le \frac{|y-y'|}{|x-y|}\cdot\frac{c_G}{w(B(x,|x-y|))}, \quad |y-y'|\le\frac{1}{2}|x-y|, \quad x,y,y'\in\Gamma.$$

In other words, K_j is a standard kernel for the measure $w(x)dx$.

This follows from Theorem 1.1 in [4].

Step 4: Kernel K_j is non-degenerate in the following sense: for every $x\in\Gamma$ and $r>0$, there exists $y\in\Gamma$ with $|x-y|=r$ and

$$|K_j(x,y)| \ge \frac{c_G}{w(B(x,|x-y|)\cap\Gamma)}.$$

Let

$$c_\Gamma = (\max_{z\in\Gamma,\ |z|=1} |z_j|)^{-1}.$$

Choose $z \in \Gamma$ such that $|z_j| = c_\Gamma^{-1}$ and $|z| = 1$. Set $y = x + rz$. Since $x, z \in \Gamma$, it follows that $y \in \Gamma$. Clearly, $|(x-y)_j| \geq rc_\Gamma^{-1}$ and $|x-y| = r$. The claim follows from the proof of the Theorem 1.2 in [4].

Step 4: Lebesgue measure on Γ is, obviously, N-regular and possesses the $(1,1)$ Poincare inequality. □

Proof of Theorem 5.1. This follows immediately from Lemma 5.4 and Theorem 1.4 in [6]. □

6. Useful part of Zhijie argument

Theorem 6.1. *If $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in B^p(\mathbb{R}^N, w(x)dx)$.*

Lemma 6.2. *Let $Q \in \mathcal{D}$. We have*

$$K_j(x,y) \geq \frac{c_k}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

Proof. TO BE DONE. Follows from Lemma 2.4.

□

Lemma 6.3. *Let real-valued $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N)$ be such that the commutator $[R_{\mathrm{Dunkl},j}, M_f]$ is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. For every $Q \in \mathcal{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$ and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_{3Q}| w(y)dy \leq c_k |\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_Q, \eta_Q\rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Proof. Let α be the median of f on $3Q$ and β be the median of f on $Q + 3s(Q)e_j$. Assume for definiteness $\alpha \geq \beta$. Set

$$A_1 = \{x \in 3Q : \ f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3s(Q)e_j : \ f(x) \leq \beta\},$$
$$A_2 = \{x \in 3Q : \ f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3s(Q)e_j : \ f(x) \geq \beta\},$$
$$\xi_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{A_s}, \quad \eta_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2.$$

We have

$$\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_{Q,s}, \eta_{Q,s}\rangle_{L_2(\mathbb{R}^N, w(x)dx)} =$$
$$= \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x,y)(f(y) - f(x))w(x)w(y)dxdy.$$

If $x \in B_1$ and $y \in A_1$, then

$$f(y) - f(x) = (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) =$$
$$= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0.$$

If $x \in B_2$ and $y \in A_2$, then

$$f(y) - f(x) = (f(y) - \beta) + (\beta - f(x)) =$$
$$= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0.$$

Thus,

$$\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_{Q,1}, \eta_{Q,1}\rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq$$
$$\geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x,y)\big(|f(y) - \alpha| + |\alpha - \beta|\big)w(x)w(y)dxdy \geq$$
$$\geq \frac{c_k}{w^2(Q)} \int_{B_1 \times A_1} \big(|f(y) - \alpha| + |\alpha - \beta|\big)w(x)w(y)dxdy =$$

$$= \frac{c_k}{w^2(Q)} \cdot \Big(w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \Big) \geq$$
$$= \frac{c_k}{w^2(Q)} \cdot \Big(\frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \Big) \geq$$
$$\geq \frac{c_k}{4w(Q)} \cdot \Big(\int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \Big)$$

and

$$-\langle [R_{\mathrm{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq$$
$$\geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x,y) |f(y) - \beta| w(x) w(y) dx dy \geq$$
$$\geq \frac{c_k}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy =$$
$$= \frac{c_k}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq$$
$$= \frac{c_k}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy.$$

It remains to note that

$$\int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy +$$
$$+ \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq$$
$$\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy$$
$$+ \int_{A_2} |\alpha - \beta| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| w(y) dy \leq$$
$$\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy + 2|\alpha - \beta| w(3Q).$$

Combining the last 3 equations, we conclude

$$\frac{c_k'}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq \max_{s=1,2} |\langle [R_{\mathrm{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$$|\langle [R_{\mathrm{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\mathrm{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\frac{c_k'}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq |\langle [R_{\mathrm{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Hence,

$$c_k' |f_{3Q} - \alpha| \leq |\langle [R_{\mathrm{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|$$

and, finally,

$$\frac{c_k'}{w(Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq 2 |\langle [R_{\mathrm{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

□

Lemma 6.4. *Suppose that $1 < p < \infty$ and $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N)$. If $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ for some $j \in \{1, 2, \cdots, N\}$, then*

$$\Big\{\frac{1}{w(3Q)}\int_{3Q}|f(x)-f_{3Q}|w(x)dx\Big\}_{Q\in\mathcal{D}} \in l_p.$$

Proof. Let $\{\xi_Q\}_{Q\in\mathcal{D}}$ and $\{\eta_Q\}_{Q\in\mathcal{D}}$ be functions constructed in Lemma 6.3. By Lemma 2.6,

$$\Big\{\langle [R_{\mathrm{Dunkl},j}, M_f]\xi_Q, \eta_Q\rangle_{L_2(\mathbb{R}^N, w(x)dx)}\Big\}_{Q\in\mathcal{D}} \in l_p.$$

The assertion follows now from Lemma 6.3. □

Lemma 6.5. *Suppose that $1 \le p < \infty$ and $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N, w(x)dx)$. We have*

$$\|f\|^p_{B^p(\mathbb{R}^N, w(x)dx)} \le c_{p,k}\sum_{Q\in\mathcal{D}}\frac{1}{w(Q)}\int_Q|f(x)-f_Q|^p w(x)dx+$$

$$+c_{p,k}\sum_{Q\in\mathcal{D}}\Big(\frac{1}{w(3Q)}\int_{3Q}|f(x)-f_{3Q}|w(x)dx\Big)^p.$$

Proof. We first decompose $\mathbb{R}^N \times \mathbb{R}^N$ into dyadic shells as follows.

$$\int_{\mathbb{R}^N\times\mathbb{R}^N}\frac{|f(x)-f(y)|^p}{w(B(x,|x-y|))^2}w(x)w(y)dxdy \le$$

$$\le \sum_{m\in\mathbb{Z}}\int_{\substack{x,y\in\mathbb{R}^N\\ 2^{-m-1}<|x-y|<2^{-m}}}\frac{|f(x)-f(y)|^p}{w(B(x,|x-y|))^2}w(x)w(y)dxdy \le$$

$$\le \sum_{m\in\mathbb{Z}}\frac{1}{w(B(x,2^{-m-1}))^2}\int_{\substack{x,y\in\mathbb{R}^N\\ |x-y|<2^{-m}}}|f(x)-f(y)|^p w(x)w(y)dxdy.$$

For $x, y \in \mathbb{R}^N$ with $2^{-m-1} < |x-y| < 2^{-m}$, fix $Q \in \mathcal{D}$ with $x \in Q$ and $s(Q) = 2^{-m}$. We have $|x-y|_\infty < 2^{-m}$ and $|x - c_Q|_\infty < 2^{-m-1}$. Thus, $|y - c_Q|_\infty < \frac{3}{2}\cdot 2^{-m}$. In other words, $y \in 3Q$. Consequently, there exists $Q \in \mathcal{D}$ with $s(Q) = 2^{-m}$, $x, y \in 3Q$. Thus,

$$\sum_{m\in\mathbb{Z}}\frac{1}{w(B(x,2^{-m-1}))^2}\int_{\substack{x,y\in\mathbb{R}^N\\ |x-y|<2^{-m}}}|f(x)-f(y)|^p w(x)w(y)dxdy \le$$

$$\le \sum_{m\in\mathbb{Z}}\frac{1}{w(B(x,2^{-m-1}))^2}\sum_{\substack{Q\in\mathcal{D}\\ s(Q)=2^{-m}}}\int_{3Q\times 3Q}|f(x)-f(y)|^p w(x)w(y)dxdy \le$$

$$\le 2^p\sum_{m\in\mathbb{Z}}\frac{1}{w(B(x,2^{-m-1}))^2}\sum_{\substack{Q\in\mathcal{D}\\ s(Q)=2^{-m}}}\int_{3Q\times 3Q}|f(x)-f_{3Q}|^p w(x)w(y)dxdy+$$

$$+2^p\sum_{m\in\mathbb{Z}}\frac{1}{w(B(x,2^{-m-1}))^2}\sum_{\substack{Q\in\mathcal{D}\\ s(Q)=2^{-m}}}\int_{3Q\times 3Q}|f(y)-f_{3Q}|^p w(x)w(y)dxdy \le$$

$$\le c_{p,k}\sum_{Q\in\mathcal{D}}\frac{1}{w^2(3Q)}\int_{3Q\times 3Q}|f(x)-f_{3Q}|^p w(x)w(y)dxdy =$$

$$= c_{p,k}\sum_{Q\in\mathcal{D}}\frac{1}{w(3Q)}\int_{3Q}|f(x)-f_{3Q}|^p w(x)dx.$$

Next,

$$\int_{3Q}|f(x)-f_{3Q}|^p w(x)dx=\sum_{\epsilon\in\{-1,0,1\}^d}\int_{Q+\epsilon s(Q)}|f(x)-f_{3Q}|^p w(x)dx\le$$

$$\le 2^{p-1}\sum_{\epsilon\in\{-1,0,1\}^d}\int_{Q+\epsilon s(Q)}|f(x)-f_{Q+\epsilon s(Q)}|^p w(x)dx+$$

$$+2^{p-1}\sum_{\epsilon\in\{-1,0,1\}^d}\int_{Q+\epsilon s(Q)}|f_{Q+\epsilon s(Q)}-f_{3Q}|^p w(x)dx.$$

Obviously,

$$\int_{Q+\epsilon s(Q)}|f_{Q+\epsilon s(Q)}-f_{3Q}|^p w(x)dx=|f_{Q+\epsilon s(Q)}-f_{3Q}|^p\cdot w(Q+\epsilon s(Q))\le$$

$$\le\Big(\int_{Q+\epsilon s(Q)}|f(x)-f_{3Q}|w(x)dx\Big)^p\cdot w^{1-p}(Q+\epsilon s(Q))\le$$

$$\le\Big(\int_{3Q}|f(x)-f_{3Q}|w(x)dx\Big)^p\cdot w^{1-p}(Q+\epsilon s(Q))\le$$

$$\le c_{p,k}\Big(\int_{3Q}|f(x)-f_{3Q}|w(x)dx\Big)^p\cdot w^{1-p}(3Q).$$

Thus,

$$\int_{\mathbb{R}^N\times\mathbb{R}^N}\frac{|f(x)-f(y)|^p}{w(B(x,|x-y|))^2}w(x)w(y)dxdy\le$$

$$\le c_{p,k}\sum_{Q\in\mathcal{D}}\sum_{\epsilon\in\{-1,0,1\}^d}\frac{1}{w(Q+\epsilon s(Q))}\int_{Q+\epsilon s(Q)}|f(x)-f_{Q+\epsilon s(Q)}|^p w(x)dx+$$

$$+c_{p,k}\sum_{Q\in\mathcal{D}}\Big(\frac{1}{w(3Q)}\int_{3Q}|f(x)-f_{3Q}|w(x)dx\Big)^p.$$

This completes the proof. □

Lemma 6.6. *Suppose that $1\le p<\infty$ and $f\in L_1^{\mathrm{loc}}(\mathbb{R}^N,w(x)dx)$. We have*

$$\sum_{Q\in\mathcal{D}}\frac{1}{w(Q)}\int_Q|f(x)-f_Q|^p w(x)dx\le c_{p,k}\sum_{Q\in\mathcal{D}}|f_Q-f_{\mathrm{parent}(Q)}|^p.$$

Proof. Obviously,

$$\sum_{Q\in\mathcal{D}}\frac{1}{w(Q)}\int_Q|f(x)-f_Q|^p w(x)dx=\sum_{m\in\mathbb{Z}}\sum_{\substack{Q\in\mathcal{D}\\ s(Q)=2^{-m}}}\frac{1}{w(Q)}\int_Q|f(x)-f_Q|^p w(x)dx\le$$

$$\le c_{p,k}\sum_{m\in\mathbb{Z}}\sum_{\substack{Q\in\mathcal{D}\\ s(Q)=2^{-m}}}\int_Q\frac{|f(x)-f_Q|^p w(x)dx}{w(B(x,2^{-m}))}=$$

$$=c_{p,k}\sum_{m\in\mathbb{Z}}\int_{\mathbb{R}^N}\frac{|f(x)-(E_mf)(x)|^p w(x)dx}{w(B(x,2^{-m}))}.$$

Note that

$$f-E_mf=\sum_{l\ge m}(E_{l+1}f-E_lf).$$

By triangle inequality,

$$\Big(\int_{\mathbb{R}^N} \frac{|f(x)-(E_m f)(x)|^p w(x)dx}{w(B(x,2^{-m}))}\Big)^{\frac{1}{p}} \leq$$
$$\leq \sum_{l\geq m} \Big(\int_{\mathbb{R}^N} \frac{|(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx}{w(B(x,2^{-m}))}\Big)^{\frac{1}{p}}.$$

Note that

$$w(B(x,2^{-m})) \geq c_k 2^{N(l-m)} w(B(x,2^{-l})).$$

Thus,

$$\Big(\int_{\mathbb{R}^N} \frac{|f(x)-(E_m f)(x)|^p w(x)dx}{w(B(x,2^{-m}))}\Big)^{\frac{1}{p}} \leq$$
$$\leq c_{p,k} \sum_{l\geq m} 2^{\frac{N(m-l)}{p}} \Big(\int_{\mathbb{R}^N} \frac{|(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx}{w(B(x,2^{-l}))}\Big)^{\frac{1}{p}}$$

and, therefore,

$$\sum_{m\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|f(x)-(E_m f)(x)|^p w(x)dx}{w(B(x,2^{-m}))} \leq$$
$$\leq c_{p,k} \sum_{m\in\mathbb{Z}} \Big(\sum_{l\geq m} 2^{\frac{N(m-l)}{p}} \Big(\int_{\mathbb{R}^N} \frac{|(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx}{w(B(x,2^{-l}))}\Big)^{\frac{1}{p}}\Big)^p.$$

Using the discreet Young's inequality

$$\|a*b\|_p \leq \|a\|_1 \|b\|_p$$

with

$$a(l) = 2^{\frac{Nl}{p}} \chi_{\mathbb{Z}_-}(l), \quad b(l) = \Big(\int_{\mathbb{R}^N} \frac{|(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx}{w(B(x,2^{-l}))}\Big)^{\frac{1}{p}},$$

we obtain

$$\sum_{m\in\mathbb{Z}} \Big(\sum_{l\geq m} 2^{\frac{N(m-l)}{p}} \Big(\int_{\mathbb{R}^N} \frac{|(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx}{w(B(x,2^{-l}))}\Big)^{\frac{1}{p}}\Big)^p \leq$$
$$\leq c_{N,p} \sum_{l\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx}{w(B(x,2^{-l}))}.$$

Thus,

$$\sum_{m\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|f(x)-(E_m f)(x)|^p w(x)dx}{w(B(x,2^{-m}))} \leq$$
$$\leq c_{p,k} \sum_{l\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx}{w(B(x,2^{-l}))} \leq$$
$$\leq c_{p,k} \sum_{l\in\mathbb{Z}} \sum_{\substack{Q\in\mathcal{D}\\ s(Q)=2^{-l-1}}} \frac{1}{w(Q)} \int_Q |(E_{l+1}f)(x)-(E_l f)(x)|^p w(x)dx =$$
$$= c_{p,k} \sum_{l\in\mathbb{Z}} \sum_{\substack{Q\in\mathcal{D}\\ s(Q)=2^{-l-1}}} |f_Q - f_{\mathrm{parent}(Q)}|^p.$$

This completes the proof. □

In what follows, conditional expectations are with respect to the measure $w(x)dx$.

Lemma 6.7. *Suppose that $1 \le p < \infty$ and $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N, w(x)dx)$. We have*

$$\|f\|^p_{B^p(\mathbb{R}^N, w(x)dx)} \le c_{p,k} \sum_{Q \in \mathcal{D}} \Big(\frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x)dx \Big)^p.$$

Proof. Combining Lemmas 6.5 and 6.6, we arrive at

$$\|f\|^p_{B^p(\mathbb{R}^N, w(x)dx)} \le c_{p,k} \sum_{Q \in \mathcal{D}} \frac{1}{w(Q)} \int_Q |f(x) - f_Q|^p w(x)dx +$$

$$+ c_{p,k} \sum_{Q \in \mathcal{D}} |f_Q - f_{\mathrm{parent(Q)}}|^p.$$

Obviously,

$$|f_Q - f_{\mathrm{parent}(Q)}| \le \frac{1}{w(Q)} \int_{\mathrm{parent}(Q)} |f(x) - f_Q| w(x)dx \le$$

$$\le \frac{1}{w(Q)} \int_{3Q} |f(x) - f_Q| w(x)dx \le \frac{c_k}{w(3Q)} \int_{3Q} |f(x) - f_Q| w(x)dx \le$$

$$\le \frac{2c_k}{w(3Q)} \int_{3Q} |f(x) - f_Q| w(x)dx.$$

This completes the proof. □

Proof of Theorem 6.1. The assertion follows by combining Lemma 6.4 and Lemma 6.7. □

7. Proof of Theorem 1.1 (ii)

Theorem 7.1. *Let $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N)$ be such that*

(i) *for every chamber Γ, we have $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$;*
(ii) *f belongs to the space $B^p(\mathbb{R}^N, w(x)dx)$;*

It follows that $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$.

Lemma 7.2. *Denote $H_+ = \mathbb{R}^{N-1} \times \mathbb{R}_+$ and $H_- = \mathbb{R}^{N-1} \times \mathbb{R}_-$. Let $z \in \mathbb{R}^N$ be such that $z_N = 0$. Suppose $f \in \dot{W}^{\frac{N}{p},p}(H_+ \cap B(z, 2\epsilon))$ and $f \in \dot{W}^{\frac{N}{p},p}(H_- \cap B(z, 2\epsilon))$. If*

$$\int_{\substack{x \in H_+ \cap B(z,2\epsilon) \\ y \in H_- \cap B(z,2\epsilon)}} \frac{|f(x) - f(y)|^p}{|x-y|^{2N+2K}} |x_N|^K |y_N|^K dxdy < \infty,$$

then

$$\int_{B(z,\epsilon)} \frac{|f(x_1, \cdots, x_{N-1}, x_N) - f(x_1, \cdots, x_{N-1}, -x_N)|^p dx}{|x_N|^N} < \infty.$$

Proof. Let $f_1(x) = f(x_1, \cdots, x_{N-1}, |x_N|)$, $x \in B(z, 2\epsilon)$. It is immediate we, probably, need reference that $f_1 \in \dot{W}^{\frac{N}{p},p}(B(z, 2\epsilon))$. In other words,

$$\int_{B(z,2\epsilon) \times B(z,2\epsilon)} \frac{|f_1(x) - f_1(y)|^p}{|x-y|^{2N}} dxdy < \infty.$$

In particular,

$$\int_{\substack{x \in H_+ \cap B(z,2\epsilon) \\ y \in H_- \cap B(z,2\epsilon)}} \frac{|f_1(x) - f_1(y)|^p}{|x-y|^{2N}} dxdy < \infty.$$

Since $|x_N|, |y_N| \le |x-y|$ for $x \in H_+$, $y \in H_-$, it follows that

$$\int_{\substack{x\in H_+\cap B(z,2\epsilon)\\ y\in H_-\cap B(z,2\epsilon)}} \frac{|f_1(x)-f_1(y)|^p}{|x-y|^{2N+2K}}|x_N|^K|y_N|^K dxdy < \infty.$$

Set $f_2 = f - f_1$. By triangle inequality,

$$\int_{\substack{x\in H_+\cap B(z,2\epsilon)\\ y\in H_-\cap B(z,2\epsilon)}} \frac{|f_2(x)-f_2(y)|^p}{|x-y|^{2N+2K}}|x_N|^K|y_N|^K dxdy < \infty.$$

Note that $f_1 = f$ on $H_+ \cap B(z, 2\epsilon)$ and, therefore, $f_2 = 0$ on $H_+ \cap B(z, 2\epsilon)$. We, therefore, have

$$\int_{\substack{x\in H_+\cap B(z,2\epsilon)\\ y\in H_-\cap B(z,2\epsilon)}} \frac{|f_2(y)|^p}{|x-y|^{2N+2K}}|x_N|^K|y_N|^K dxdy < \infty.$$

In other words,

$$\int_{H_-\cap B(z,2\epsilon)} |f_2|^p(y)\Big(\int_{H_+\cap B(z,2\epsilon)} \frac{|x_N|^K|y_N|^K dx}{|x-y|^{2N+2K}}\Big)dy < \infty.$$

If $y \in B(z, \epsilon)$, then $B(y', \epsilon) \subset B(z, 2\epsilon)$. Thus,

$$\int_{H_-\cap B(z,\epsilon)} |f_2|^p(y)\Big(\int_{H_+\cap B(y',\epsilon)} \frac{|x_N|^K|y_N|^K dx}{|x-y|^{2N+2K}}\Big)dy < \infty.$$

Clearly,

$$\int_{H_+\cap B(y',\epsilon)} \frac{|x_N|^K|y_N|^K dx}{|x-y|^{2N+2K}} = \int_{H_+\cap B(0,\epsilon)} \frac{|x_N|^K|y_N|^K dx}{(|x'|^2+(x_N+|y_N|)^2)^{N+K}}.$$

Thus,

$$\int_{H_-\cap B(z,\epsilon)} |f_2|^p(y)\Big(\int_{H_+\cap B(0,\epsilon)} \frac{|x_N|^K|y_N|^K dx}{\max\{|x'|, x_N, |y_N|\}^{2N+2K}}\Big)dy < \infty.$$

Clearly,

$$\int_{H_+\cap B(0,\epsilon)} \frac{|x_N|^K|y_N|^K dx}{\max\{|x'|, x_N, |y_N|\}^{2N+2K}} \ge \int_{\substack{|x'|<\frac12|y_N|\\ x_N\in(0,\frac12|y_N|)}} \frac{|x_N|^K|y_N|^K dx}{|y_N|^{2N+2K}} =$$

$$= |y_N|^{-2N-K}\cdot\int_{|x'|<\frac12|y_N|} dx'\cdot\int_0^{\frac12|y_N|} dx_N = |y_N|^{-N}\cdot 2^{-N}\mathrm{Vol}(\mathbb{B}^{N-1}).$$

Hence,

$$\int_{H_-\cap B(z,\epsilon)} \frac{|f_2|^p(y)dy}{|y_N|^N} < \infty.$$

This completes the proof. □

Lemma 7.3. *Denote $H_+ = \mathbb{R}^{N-1}\times\mathbb{R}_+$ and $H_- = \mathbb{R}^{N-1}\times\mathbb{R}_-$. Let $z \in \mathbb{R}^N$ be such that $z_N = 0$. Suppose $f \in \dot{W}^{\frac{N}{p},p}(H_+\cap B(z,\epsilon))$ and $f \in \dot{W}^{\frac{N}{p},p}(H_-\cap B(z,\epsilon))$. If*

$$\int_{B(z,\epsilon)} \frac{|f(x_1,\cdots,x_{N-1},x_N)-f(x_1,\cdots,x_{N-1},-x_N)|^p dx}{|x_N|^N} < \infty,$$

then $f \in \dot{W}^{\frac{N}{p},p}(B(z,\epsilon))$.

Proof. Set $r(x) = (x_1, \cdots, x_{N-1}, -x_N)$, $x \in B(z,\epsilon)$ and let $f_1 = f + f \circ r$, $f_2 = f - f \circ r$. It is known that provide reference that $f_1 \in \dot{W}^{\frac{N}{p},p}(B(z,\epsilon))$. Hence, we may assume without loss of generality that $f = -f \circ r$ and

$$\int_{B(z,\epsilon)} \frac{|f|^p(x)dx}{|x_N|^N} < \infty.$$

We have

$$\int_{B(z,\epsilon)\times B(z,\epsilon)} \frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}} \leq$$
$$\leq \int_{\substack{x\in H_+\cap B(z,\epsilon)\\ y\in H_+\cap B(z,\epsilon)}} \frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}} + \int_{\substack{x\in H_-\cap B(z,\epsilon)\\ y\in H_-\cap B(z,\epsilon)}} \frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}} +$$
$$+2^{p-1}\int_{\substack{x\in H_+\cap B(z,\epsilon)\\ y\in H_-\cap B(z,\epsilon)}} \frac{|f|^p(x)dxdy}{|x-y|^{2N}} + 2^{p-1}\int_{\substack{x\in H_+\cap B(z,\epsilon)\\ y\in H_-\cap B(z,\epsilon)}} \frac{|f|^p(y)dxdy}{|x-y|^{2N}} +$$
$$+2^{p-1}\int_{\substack{x\in H_-\cap B(z,\epsilon)\\ y\in H_+\cap B(z,\epsilon)}} \frac{|f(x)|^p dxdy}{|x-y|^{2N}} + 2^{p-1}\int_{\substack{x\in H_-\cap B(z,\epsilon)\\ y\in H_+\cap B(z,\epsilon)}} \frac{|f|^p(y)dxdy}{|x-y|^{2N}}.$$

It remains to note that, for every $x \in H_+$,

$$\int_{y\in H_-\cap B(z,\epsilon)} \frac{dy}{|x-y|^{2N}} \leq \int_{y\in\mathbb{R}^N} \frac{dy}{(|y|'^2 + (|y_N| + |x_N|)^2)^N} = c_N|x_N|^{-N}.$$

Thus,

$$\int_{B(z,\epsilon)\times B(z,\epsilon)} \frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}} \leq$$
$$\leq \int_{\substack{x\in H_+\cap B(z,\epsilon)\\ y\in H_+\cap B(z,\epsilon)}} \frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}} + \int_{\substack{x\in H_-\cap B(z,\epsilon)\\ y\in H_-\cap B(z,\epsilon)}} \frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}} +$$
$$+2^{p-1}c_N\int_{x\in H_+\cap B(z,\epsilon)} \frac{|f|^p(x)dx}{|x_N|^N} + 2^{p-1}c_N\int_{H_-\cap B(z,\epsilon)} \frac{|f|^p(y)dy}{|y_N|^N} +$$
$$+2^{p-1}c_N\int_{H_-\cap B(z,\epsilon)} \frac{|f(x)|^p dx}{|x_N|^N} + 2^{p-1}c_N\int_{H_+\cap B(z,\epsilon)} \frac{|f|^p(y)dy}{|y_N|^N}.$$

Since the right hand side is finite, the assertion follows. □

Lemma 7.4. *Let Γ_1 and Γ_2 be distinct chambers sharing the same face $P \subset \{\langle x, v\rangle = 0\}$. Let z be in the relative interior of P. If f is as in Theorem 7.1, then $f \in \dot{W}^{\frac{N}{p},p}(B(z,\epsilon))$ for every sufficiently small $\epsilon > 0$.*

Proof. To simplify the notations, let $v = e_N$. Denote for brevity, $k_{e_N} = K$. Let z

For every $\epsilon > 0$, we have

$$\int_{B(z,2\epsilon)\times B(z,2\epsilon)} \frac{|f(x)-f(y)|^p}{w(B(x,|x-y|))^2} w(x)w(y)dxdy < \infty.$$

Now, let $\epsilon > 0$ be so small that $|\langle x, \tilde{v}\rangle| \geq c_{z,\epsilon} > 0$ for every $v \neq \tilde{v} \in R$ and for every $x \in B(z, 2\epsilon)$. It follows that

$$w(x) \geq c_{z,\epsilon,k}|x_N|^K, \quad w(y) \geq c_{z,\epsilon,k}|y_N|^K, \quad x, y \in B(z, 2\epsilon).$$

Also,

$$w(B(x,|x-y|)) \leq c_{z,\epsilon,k}|x-y|^N \cdot (|x-y| + |x_N|)^K, \quad x, y \in B(z, 2\epsilon).$$

Consequently,

$$w(B(x,|x-y|)) \le c_{z,\epsilon,k}|x-y|^{N+K}, \quad x \in H_+ \cap B(z,2\epsilon), \quad y \in H_- \cap B(z,2\epsilon).$$

We, therefore, have

$$\int_{\substack{x\in H_+\cap B(z,2\epsilon)\\ y\in H_-\cap B(z,2\epsilon)}} \frac{|f(x)-f(y)|^p}{|x-y|^{2N+2K}}|x_N|^K|y_N|^K dxdy < \infty.$$

On the other hand, $H_+ \cap B(z,2\epsilon) \subset \Gamma_1$ and $H_- \cap B(z,2\epsilon) \subset \Gamma_2$. Thus, $f \in \dot{W}^{\frac{N}{p},p}(H_+ \cap B(z,2\epsilon))$ and $f \in \dot{W}^{\frac{N}{p},p}(H_- \cap B(z,2\epsilon))$. By Lemma 7.2 and Lemma 7.3, $f \in \dot{W}^{\frac{N}{p},p}(B(z,\epsilon))$. □

Lemma 7.5. *Suppose $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$ for every chamber Γ. Suppose that, for every face of every chamber and for every z in the relative interior of that face, we have $f \in \dot{W}^{\frac{N}{p},p}(B(z,\epsilon))$ for every sufficiently small $\epsilon > 0$. It follows that $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$. Furthermore, we have*

$$\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)} \le c_R \sum_{\Gamma} \|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)}.$$

Proof. find reference. □

Proof of Theorem 7.1. The assertion follows by combining Lemmas 7.4 and 7.5. □

Proof of Theorem 1.1 (ii). The assertion follows by combining Theorem 5.1, Theorem 6.1 and Theorem 7.1. □

8. Constant characterization

Theorem 8.1. *If $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N, w(x)dx)$ is such that*

$$[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$$

for some $p \le d$, then f is constant.

Proof. By Lemma 5.4, measures dx and $w(x)dx$ and operator $T_{\Gamma,j} = M_{\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_\Gamma}$ satisfy on Γ the assumptions in Theorem 1.4 in [6]. By assumption, $[T_{\Gamma,j}, M_f] \in \mathcal{L}_p(L_2(\Gamma, w(x)dx))$. By Theorem 1.4 in [6], f is constant on Γ.

Hence, f is constant on every chamber. Let $f = c_1$ on Γ_1 and $f = c_2$ on Γ_2. On the other hand, $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{2N}(L_2(\mathbb{R}^N, w(x)dx))$. By Theorem 1.1 (ii), $f \in \dot{W}^{\frac{1}{2},2N}(\mathbb{R}^N)$. We have

$$\int_{\Gamma_1\times\Gamma_2} \frac{|c_1-c_2|^{2N}dxdy}{|x-y|^{2N}} = \int_{\Gamma_1\times\Gamma_2} \frac{|f(x)-f(y)|^{2N}dxdy}{|x-y|^{2N}} \le$$

$$\le \int_{\mathbb{R}^N\times\mathbb{R}^N} \frac{|f(x)-f(y)|^{2N}dxdy}{|x-y|^{2N}} = \|f\|_{\dot{W}^{\frac{1}{2},2N}(\mathbb{R}^N)} < \infty.$$

It follows that $c_1 = c_2$. Since chambers Γ_1 and Γ_2 are arbitrary, the assertion follows. □

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